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GATE ECE · NETWORK THEORY · CHAPTER-WISE PYQS

Two Port Networks

GATE ECE Network Theory · Previous Year Questions with Answer Key

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Marks 1	16 questions
Marks 2	34 questions
Marks 5	8 questions
Marks 8	1 question
Answer key	at the end

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Marks 1 16 questions

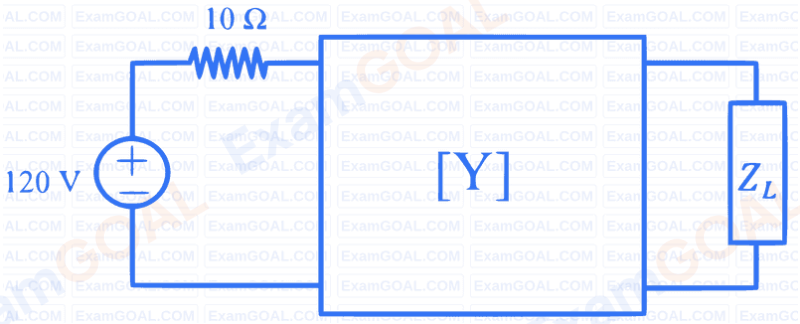
Q1. GATE ECE 2023

Open solution

For the two port network shown below, the [Y]-parameters is given as

$$[Y] = \frac{1}{100}\begin{bmatrix} 2 & -1 \\ -1 & 4/3 \end{bmatrix} S$$

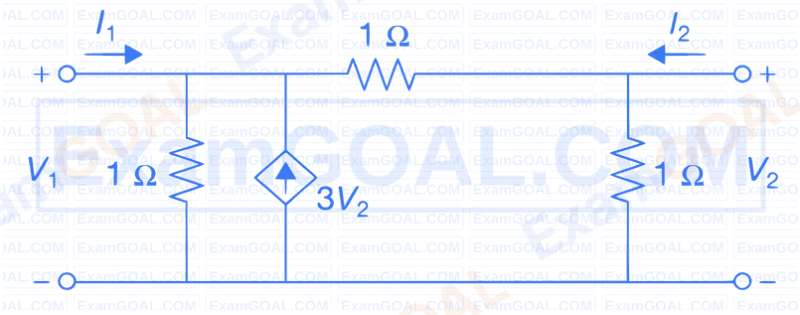
The value of load impedance Z_L, in Ω, for maximum power transfer will be _____ (rounded off to the nearest integer).



Q2. GATE ECE 2021

Open solution

Consider the two-port network shown in the figure.



The admittance parameters, in siemens are

(A) $y_{11} = 2, y_{12} = -4, y_{21} = -1, y_{22} = 2$

(B) $y_{11} = 2, y_{12} = -4, y_{21} = -4, y_{22} = 2$

(C) $y_{11} = 2, y_{12} = -4, y_{21} = -4, y_{22} = 3$

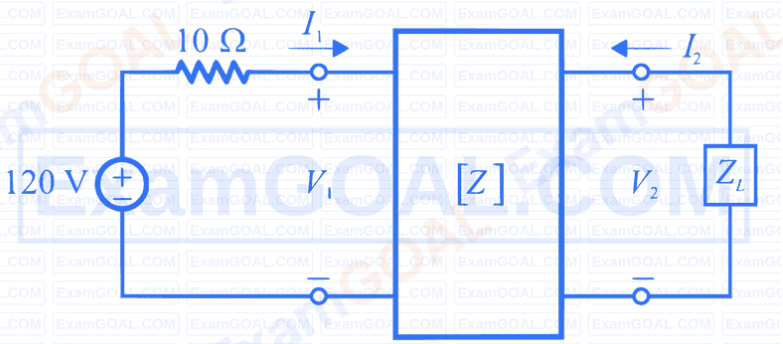
(D) $y_{11} = 1, y_{12} = -2, y_{21} = -1, y_{22} = 3$

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Q3. GATE ECE 2020

In the given circuit, the two-port network has the impedance matrix $[Z] = \begin{bmatrix} 40 & 60 \\ 60 & 120 \end{bmatrix}$. The value of Z_L for which maximum power is transferred to the load is ____ Ω.

Open solution

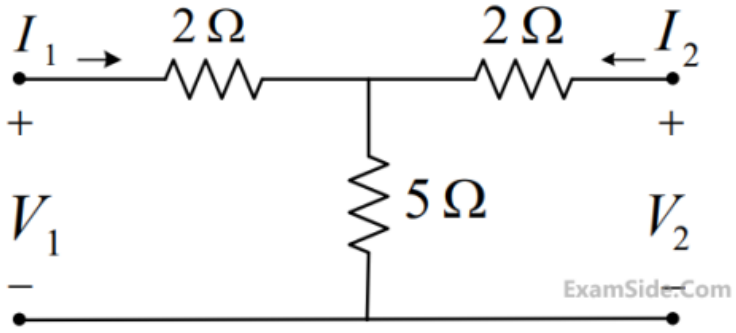


Q4. GATE ECE 2018

The ABCD matrix for a two-port network is defined by :

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Open solution



The parameter B for the given two-port network (in ohms, correct to two decimal places) is _____.

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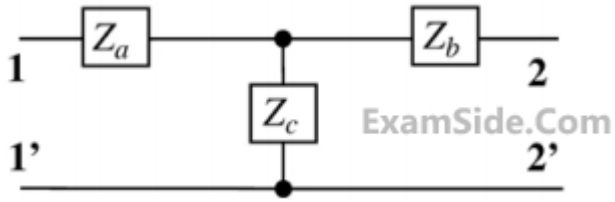
Q5. GATE ECE 2016 Set 2

Open solution

The z-parameter matrix for the two-port network shown is

$$\begin{bmatrix} 2j\omega & j\omega \\ j\omega & 3+2j\omega \end{bmatrix}$$

Where the entries are in Ω. Suppose $Z_b\ (j\omega) = R_b + j\omega$



Then the value of R_b (in Ω) equals _____.

Q6. GATE ECE 2016 Set 1

Open solution

Consider a two-port network with the transmission matrix: T = $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$. If the network is reciprocal, then

(A) $T^{-1} = T$

(B) $T^2 = T$

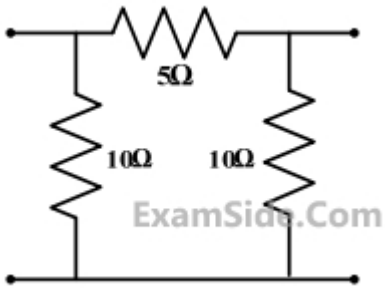
(C) Determinant (T) = 0

(D) Determinant (T) = 1

Q7. GATE ECE 2015 Set 2

Open solution

The 2-port admittance matrix of the circuit shown is given by



(A) $\begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}$

(B) $\begin{bmatrix} 15 & 5 \\ 5 & 15 \end{bmatrix}$

(C) $\begin{bmatrix} 3.33 & 5 \\ 5 & 3.33 \end{bmatrix}$

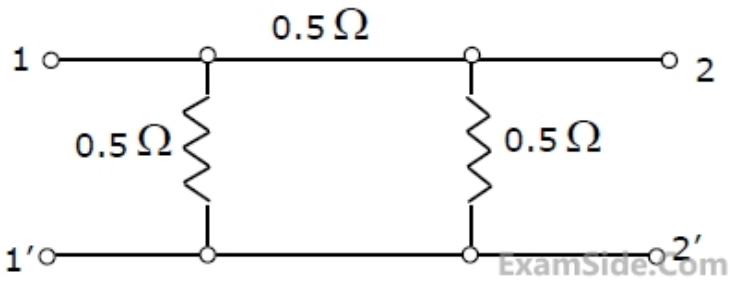
(D) $\begin{bmatrix} 0.3 & 0.4 \\ 0.4 & 0.3 \end{bmatrix}$

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Q8. GATE ECE 2010

Open solution

For the two-port network shown below, the short-circuit admittance parameter matrix is



(A) $\begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} S$

(B) $\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} S$

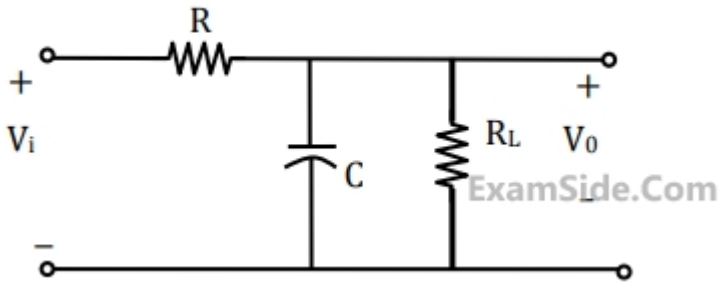
(C) $\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} S$

(D) $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} S$

Q9. GATE ECE 2009

Open solution

If the transfer function of the following network is $\frac{V_0(s)}{V_i(s)} = \frac{1}{2+sCR}$



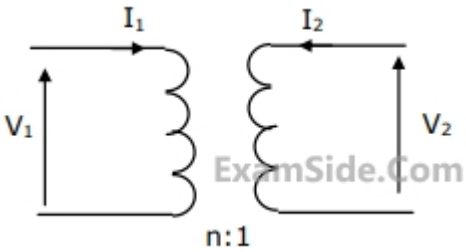
The value of the load resistance R_L is

(A) R/4 (B) R/2 (C) R (D) 2R

Q10. GATE ECE 2005

Open solution

The ABCD parameters of an ideal n:1 transformer shown in Fig. are $\begin{bmatrix} n & 0 \\ 0 & X \end{bmatrix}$. The value of X will be



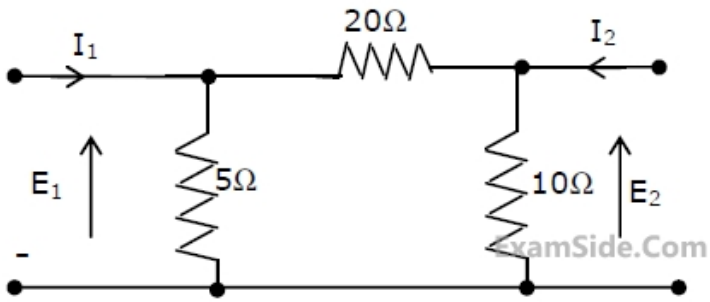
(A) n (B) 1/n (C) n^2 (D) $1/n^2$

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Q11. GATE ECE 2001

Open solution

The admittance parameter Y_{12} in the 2-port network in Figure is



(A) -0.2 mho (B) 0.1 mho (C) -0.05 mho (D) 0.05 mho

Q12. GATE ECE 2001

Open solution

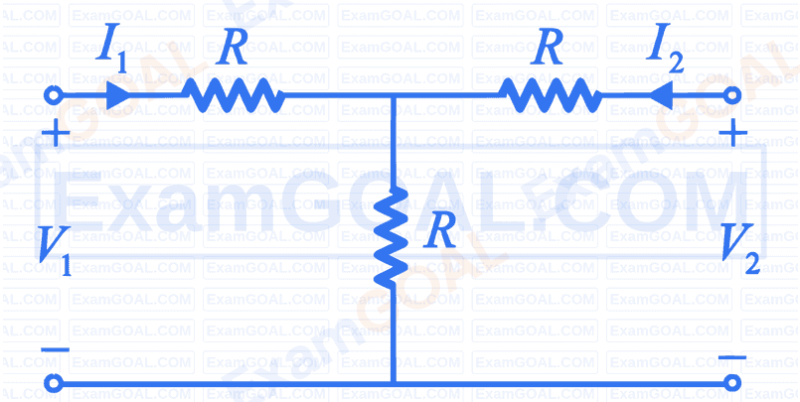
If each branch of a Delta circuit has impedance $\sqrt{3}\,Z$, then each branch of the equivalent Wye circuit has impedance

(A) $\frac{Z}{\sqrt{3}}$ (B) 3Z (C) $3\sqrt{3}\,Z$ (D) $\frac{Z}{3}$

Q13. GATE ECE 1999

Open solution

A 2-port network is shown in figure. The parameter h_{21} for this network can be given by



(A) - 1/2 (B) + 1/2 (C) - 3/2 (D) + 3/2

Q14. GATE ECE 1998

Open solution

The short-circuit admittance matrix of a two-port network is

$$\begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$$

The two port network is

(A) non-reciprocal and passive
(B) non-reciprocal and active
(C) reciprocal and passive
(D) reciprocal and active

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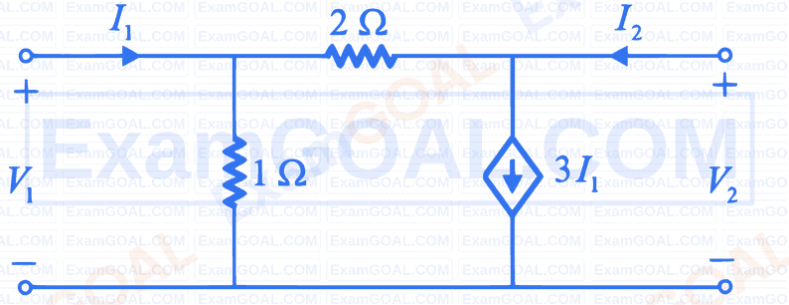
Q15. GATE ECE 1994

The condition, that a 2-port network is reciprocal, can be expressed in terms of its ABCD parameters as .....

Open solution

Q16. GATE ECE 1990

The open circuit impedance matrix of the 2-port network shown in figure is



(A) $\begin{bmatrix} -2 & 1 \\ -8 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} -2 & -8 \\ -8 & 3 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

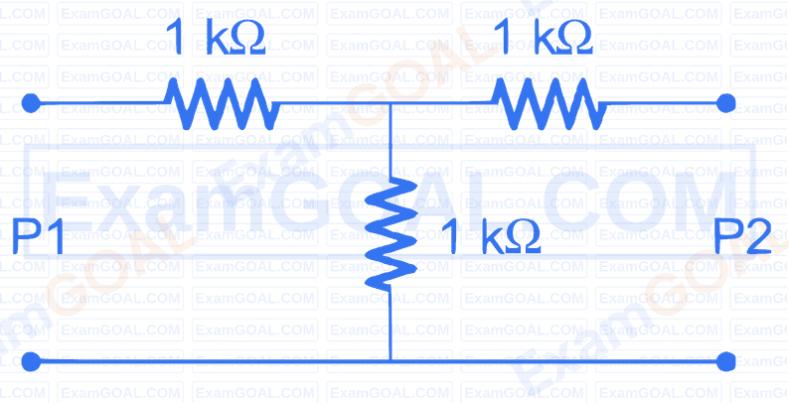
Open solution

Marks 2 34 questions

Q17. GATE ECE 2026

Consider the two-port network as shown in the Figure.

Which of the following options provides the correct set of values of A, B, C and D parameters?



(A) $A = \frac{1}{2}, B = 3 \times 10^3 \Omega, C = 10^{-3} \Omega^{-1}, D = \frac{1}{2}$

(B) $A = 2, B = 3\Omega, C = 1\Omega^{-1}, D = 2$

(C) $A = 2, B = 6 \times 10^3 \Omega, C = 2 \times 10^{-3} \Omega^{-1}, D = 2$

(D) $A = 2, B = 3 \times 10^3 \Omega, C = 10^{-3} \Omega^{-1}, D = 2$

Open solution

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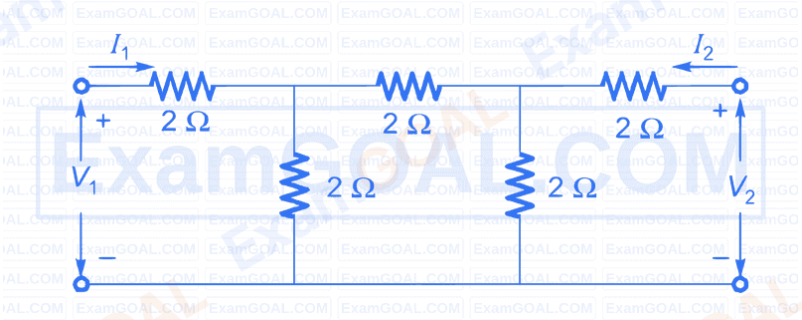
Q18. GATE ECE 2025

Open solution

The Z-parameter matrix of a two port network relates the port voltages and port currents as follows:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = Z \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

The Z-parameter matrix (with each entry in Ohms) of the network shown below is _____.



(A) $\begin{bmatrix} \frac{10}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{10}{3} \end{bmatrix}$

(B) $\begin{bmatrix} \frac{2}{3} & \frac{10}{3} \\ \frac{10}{3} & \frac{2}{3} \end{bmatrix}$

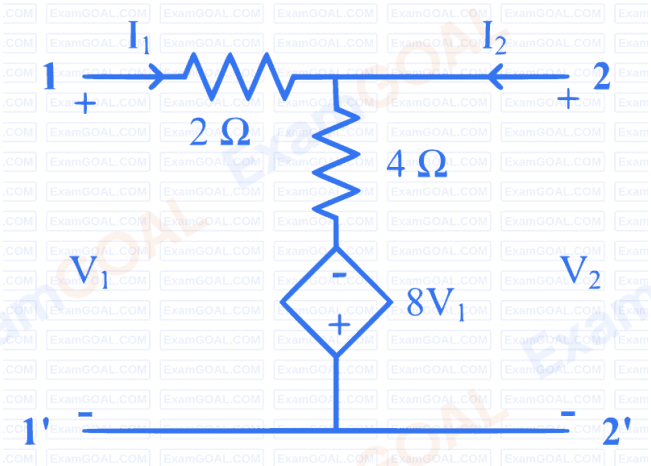
(C) $\begin{bmatrix} 10 & 2 \\ 2 & 10 \end{bmatrix}$

(D) $\begin{bmatrix} \frac{10}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{10}{3} \end{bmatrix}$

Q19. GATE ECE 2024

Open solution

For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____.

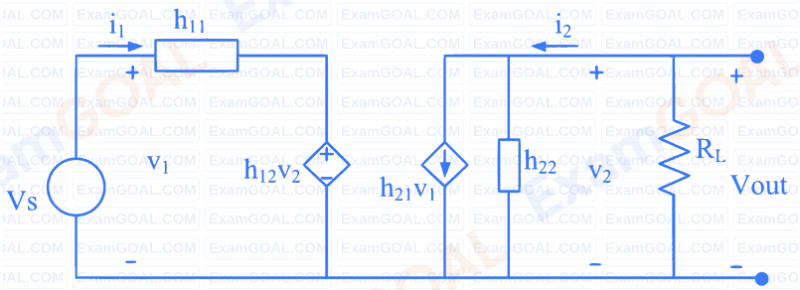


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Q20. GATE ECE 2023

Open solution

The h-parameters of a two port network are shown below. The condition for the maximum small signal voltage gain $\frac{V_{out}}{V_s}$ is



(A) $h_{11} = 0, h_{12} = 0, h_{21} =$ very high and $h_{22} = 0$

(B) $h_{11} =$ very high, $h_{12} = 0, h_{21} =$ very high and $h_{22} = 0$

(C) $h_{11} = 0, h_{12} =$ very high, $h_{21} =$ very high and $h_{22} = 0$

(D) $h_{11} = 0, h_{12} = 0, h_{21} =$ very high and $h_{22} =$ very high

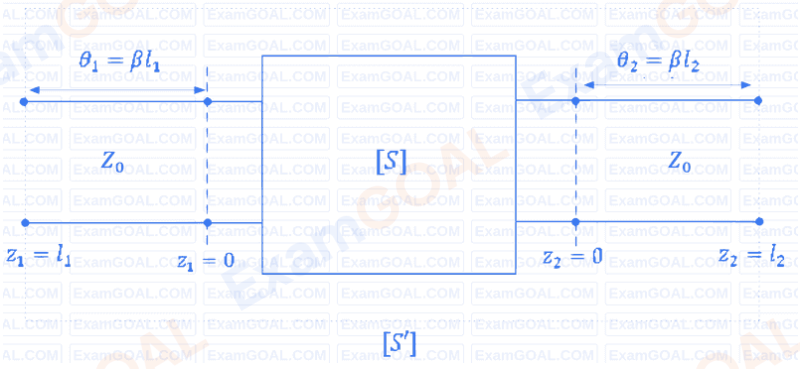
Q21. GATE ECE 2023

Open solution

The S-parameters of a two port network is given as

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

with reference to Z_0. Two lossless transmission line sections of electrical lengths $\theta_1 = \beta l_1$ and $\theta_2 = \beta l_2$ are added to the input and output ports for measurement purposes, respectively. The S-parameters $[S']$ of the resultant two port network is



(A) $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$

(B) $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$

(C) $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$

(D) $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$

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Q22. GATE ECE 2022

Open solution

A linear 2-port network is shown in Fig. (a). An ideal DC voltage source of 10 V is connected across Port 1. A variable resistance R is connected across Port 2. As R is varied, the measured voltage and current at Port 2 is shown in Fig. (b) as a V_2 versus $-I_2$ plot. Note that for V_2 = 5 V, I_2 = 0 mA, and for V_2 = 4 V, I_2 = −4 mA. When the variable resistance R at Port 2 is replaced by the load shown in Fig. (c), the current I_2 is _____ mA (rounded off to one decimal place).

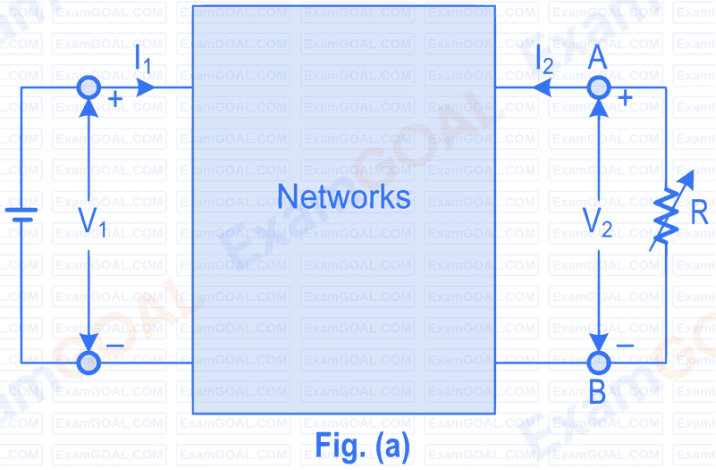


Fig. (a)

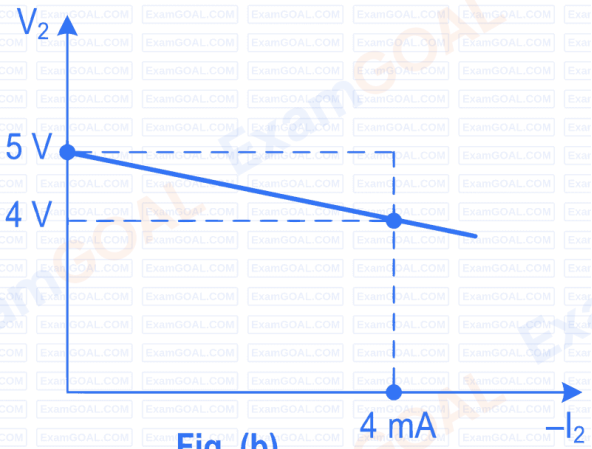


Fig. (b)

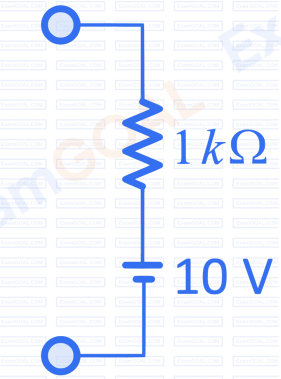


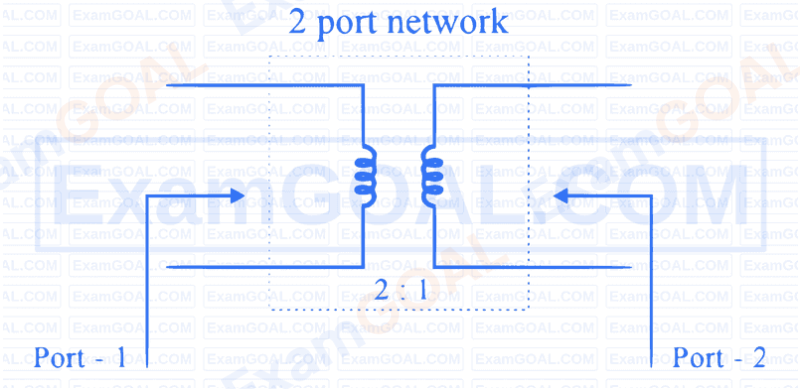
Fig. (c)

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Q23. GATE ECE 2020

For a 2-port network consisting of an ideal lossless transformer, the parameter S_{21} (rounded off to two decimal places) for a reference impedance of 10Ω is ____ .

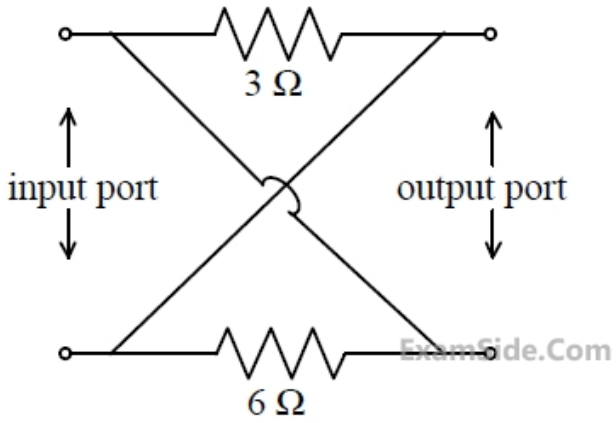
Open solution



Q24. GATE ECE 2016 Set 3

The z-parameter matrix $\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$ for the two-port network shown is

Open solution



(A) $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 9 & -3 \\ 6 & 9 \end{bmatrix}$

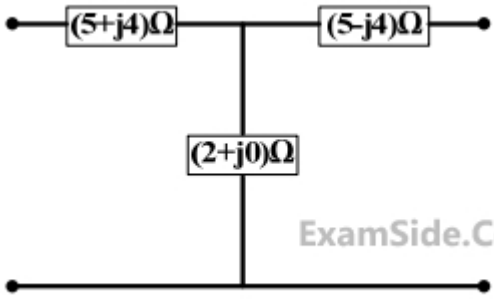
(D) $\begin{bmatrix} 9 & 3 \\ 6 & 9 \end{bmatrix}$

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Q25. GATE ECE 2015 Set 3

Open solution

The ABCD parameters of the following 2-port network are



(A) $\begin{bmatrix} 3.5 + j2 & 20.5 \\ 20.5 & 3.5 - j2 \end{bmatrix}$

(B) $\begin{bmatrix} 3.5 + j2 & 30.5 \\ 0.5 & 3.5 - j2 \end{bmatrix}$

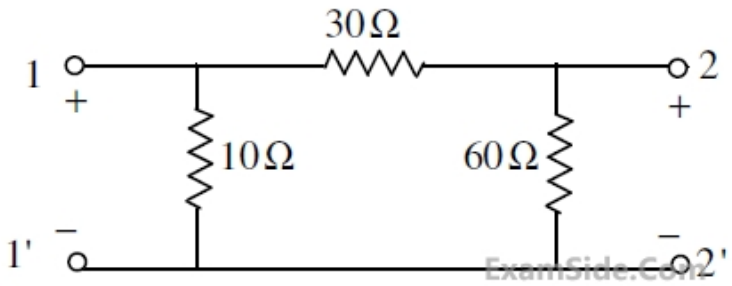
(C) $\begin{bmatrix} 10 & 2 + j0 \\ 2 + j0 & 10 \end{bmatrix}$

(D) $\begin{bmatrix} 7 + j4 & 0.5 \\ 30.5 & 7 - j4 \end{bmatrix}$

Q26. GATE ECE 2014 Set 4

Open solution

For the two-port network shown in the figure, the impedance (Z) matrix (in Ω) is



(A) $\begin{bmatrix} 6 & 24 \\ 42 & 9 \end{bmatrix}$

(B) $\begin{bmatrix} 9 & 8 \\ 8 & 24 \end{bmatrix}$

(C) $\begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$

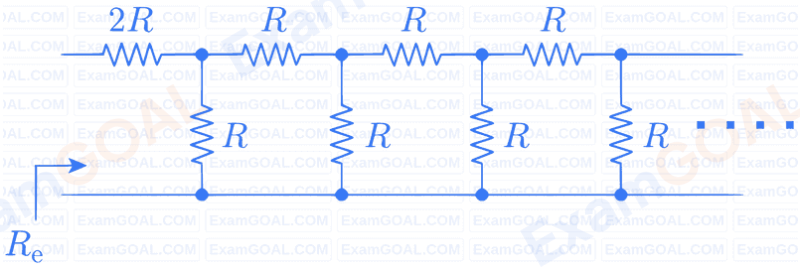
(D) $\begin{bmatrix} 42 & 6 \\ 6 & 60 \end{bmatrix}$

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Q27. GATE ECE 2014 Set 4

Open solution

The equivalent resistance in the infinite ladder network shown in the figure, is R_e

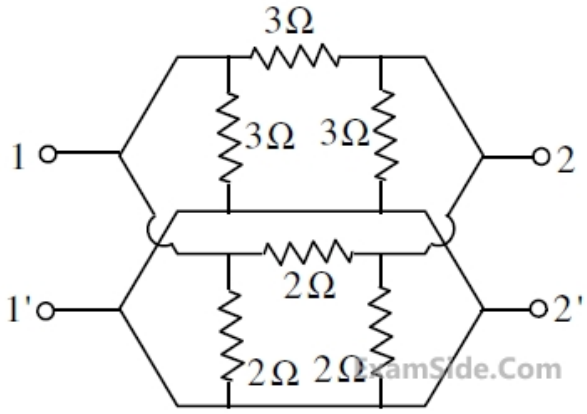


The value of R_e/R is _____

Q28. GATE ECE 2014 Set 2

Open solution

In the h-parameter model of the 2-port network given in the figure shown, the value of h_{22} (in S) is _____ .



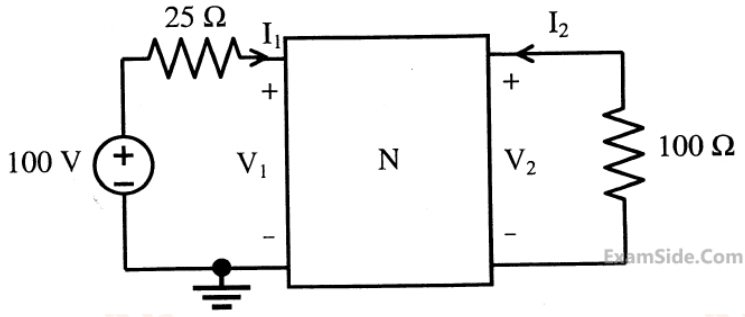
Q29. GATE ECE 2011

Open solution

In the circuit shown below, the network N is described by the following Y matrix:

$$Y = \begin{bmatrix} 0.1S & -0.01S \\ 0.01S & 0.1S \end{bmatrix}$$

The voltage gain $\frac{V_2}{V_1}$ is



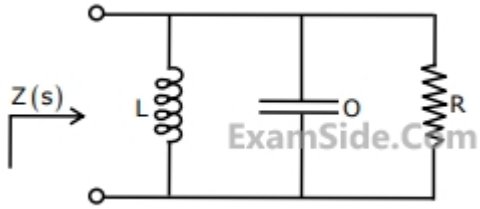
(A) 1/90 (B) -1/90 (C) -1/99 (D) -1/11

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Q30. GATE ECE 2008

The driving point impedance of the following network is given by $Z(s) = \frac{0.2\,s}{s^2 + 0.1\,s + 2}$. The component values are

Open solution



(A) L = 5H, R = 0.5 Ω, C = 0.1 F

(B) L = 0.1 H, R = 0.5 Ω, C = 5 F

(C) L = 5H, R = 2 Ω, C = 0.1 F

(D) L = 0.1 H, R = 2 Ω, C = 5 F

Q31. GATE ECE 2008

A two-port network shown below is excited by external dc sources. The voltages and the currents are measured with voltmeters V_1, V_2 and ammeter A_1, A_2 (all assumed to be ideal), as indicated. Under following switch conditions, the readings obtained are:

(i) $S_1 - open, \quad S_2 - closed$

$A_1 = 0\,A, \quad V_1 = 4.5\,V,$

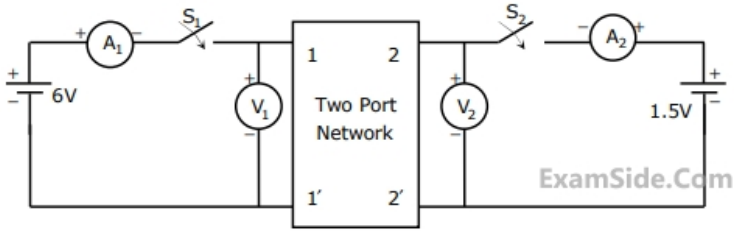
$V_2 = 1.5\,V, \quad A_2 = 1\,A$

(ii) $S_1 - Closed, \quad S_2 - Open$

$A_1 = 4\,A, \quad V_1 = 6\,V,$

$V_2 = 6\,V, \quad A_2 = 0\,A$

Open solution



The z-parameter matrix for this network is

(A) $\begin{bmatrix} 1.5 & 1.5 \\ 4.5 & 1.5 \end{bmatrix}$

(B) $\begin{bmatrix} 1.5 & 4.5 \\ 1.5 & 4.5 \end{bmatrix}$

(C) $\begin{bmatrix} 1.5 & 4.5 \\ 1.5 & 1.5 \end{bmatrix}$

(D) $\begin{bmatrix} 4.5 & 1.5 \\ 1.5 & 4.5 \end{bmatrix}$

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Q32. GATE ECE 2008

Open solution

A two-port network shown below is excited by external dc sources. The voltages and the currents are measured with voltmeters V_1, V_2 and ammeter A_1, A_2 (all assumed to be ideal), as indicated. Under following switch conditions, the readings obtained are:

(i) $S_1 - open, \quad S_2 - closed$

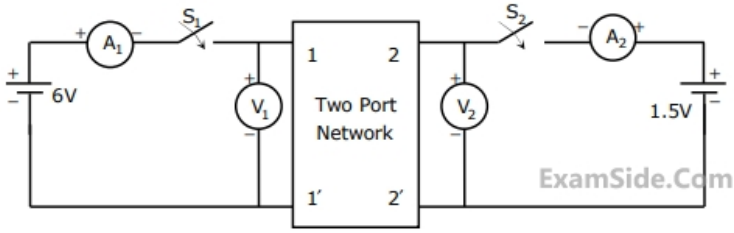
$A_1 = 0\,A, \quad V_1 = 4.5\,V,$

$V_2 = 1.5\,V, \quad A_2 = 1\,A$

(ii) $S_1 - Closed, \quad S_2 - Open$

$A_1 = 4\,A, \quad V_1 = 6\,V,$

$V_2 = 6\,V, \quad A_2 = 0\,A$



The h-parameter matrix for this network is

(A) $\begin{bmatrix} -3 & 3 \\ -1 & 0.67 \end{bmatrix}$

(B) $\begin{bmatrix} -3 & -1 \\ 3 & 0.67 \end{bmatrix}$

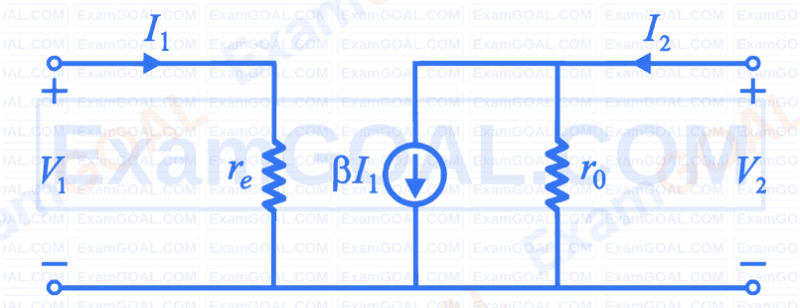
(C) $\begin{bmatrix} 3 & 3 \\ 1 & 0.67 \end{bmatrix}$

(D) $\begin{bmatrix} 3 & 1 \\ -3 & -0.67 \end{bmatrix}$

Q33. GATE ECE 2006

Open solution

In the two port network shown in the figure below, z_{12} and z_{21} are, respectively



(A) $r_0 \; and \; \beta\, r_0$ (B) $0 \; and \; -\beta\, r_0$ (C) $0 \; and \; \beta\, r_0$ (D) $r_0 \; and \; -\beta\, r_0$

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Q34. GATE ECE 2006

Open solution

A two-port network is represented by ABCD parameters given by

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

If port-2 is terminated by R_L, the input impedance seen at port-1 is given by

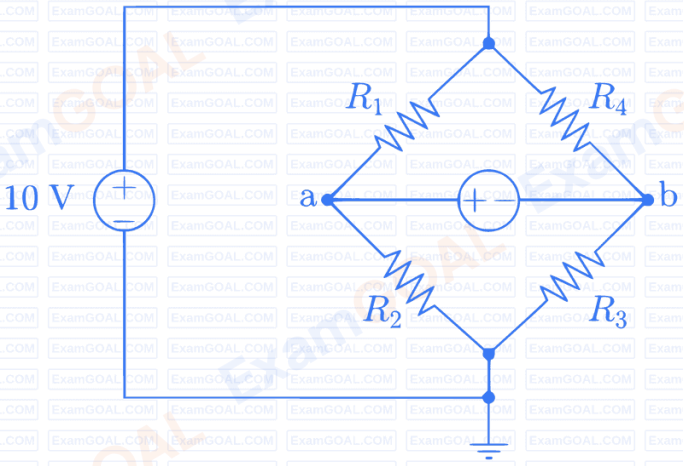
(A) $\frac{A+BR_L}{C+DR_L}$ (B) $\frac{AR_L+C}{BR_L+D}$

(C) $\frac{DR_L+A}{BR_L+C}$ (D) $\frac{B+AR_L}{D+CR_L}$

Q35. GATE ECE 2005

Open solution

If $R_1 = R_2 = R_4 = R$ and $R_3 = 1.1R$ in the bridge circuit shown in figure, then the reading in the ideal voltmeter connected between a and b is

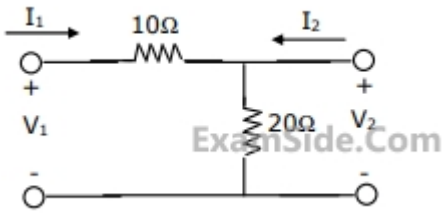


(A) 0.238 V (B) 0.138 V (C) -0.238 V (D) 1 V

Q36. GATE ECE 2005

Open solution

The h parameters of the circuit shown in Fig. are



(A) $\begin{bmatrix} 0.1 & 0.1 \\ -0.1 & 0.3 \end{bmatrix}$

(B) $\begin{bmatrix} 10 & -1 \\ 1 & 0.05 \end{bmatrix}$

(C) $\begin{bmatrix} 30 & 20 \\ 20 & 20 \end{bmatrix}$

(D) $\begin{bmatrix} 10 & 1 \\ -1 & 0.05 \end{bmatrix}$

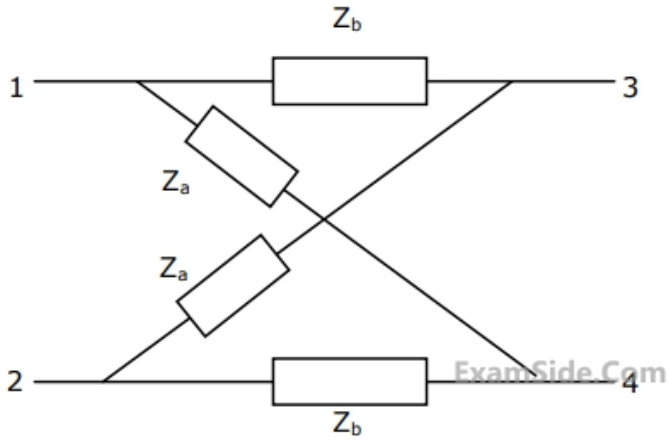
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Q37. GATE ECE 2004

Open solution

For the lattice circuit shown in Fig., $Z_a = j\,2\,\Omega$ *and* $Z_b = \ 2\Omega$. The values of the open circuit impedance parameters

$Z \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$ are



(A) $\begin{bmatrix} 1-j & 1+j \\ 1+j & 1+j \end{bmatrix}$

(B) $\begin{bmatrix} 1-j & 1+j \\ -1+j & 1-j \end{bmatrix}$

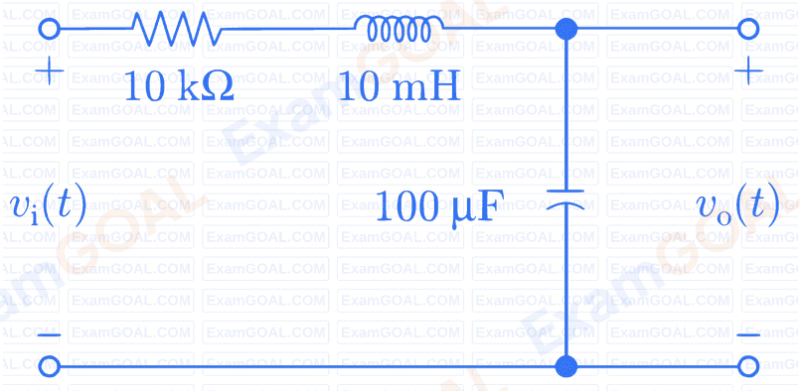
(C) $\begin{bmatrix} 1+j & 1+j \\ 1-j & 1-j \end{bmatrix}$

(D) $\begin{bmatrix} 1+j & j-1 \\ j-1 & 1+j \end{bmatrix}$

Q38. GATE ECE 2004

Open solution

For the circuit shown in Figure, the initial conditions are zeros. Its transfer function $H(s) = \frac{V_c\,(s)}{V_i\,(s)}$ is



(A) $\frac{1}{s^2 + 10^6\,s + 10^6}$

(B) $\frac{10^6}{s^2 + 10^3\,s + 10^6}$

(C) $\frac{10^3}{s^2 + 10^3\,s + 10^6}$

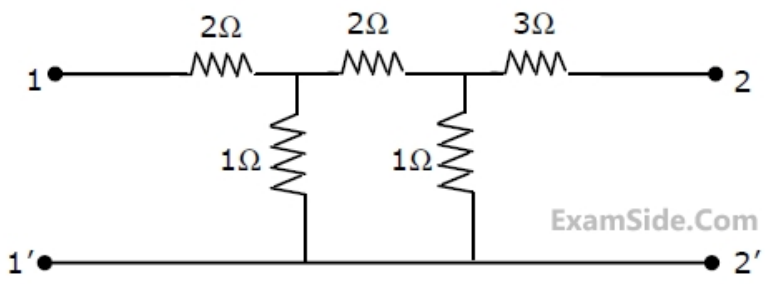
(D) $\frac{10^6}{s^2 + 10^6\,s + 10^6}$

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Q39. GATE ECE 2003

Open solution

The impedance parameters Z_{11} and Z_{12} of the two-port network in figure are



(A) Z_{11} = 2.75Ω and Z_{12} =0.25Ω

(B) Z_{11} = 3Ω and Z_{12} =0.5Ω

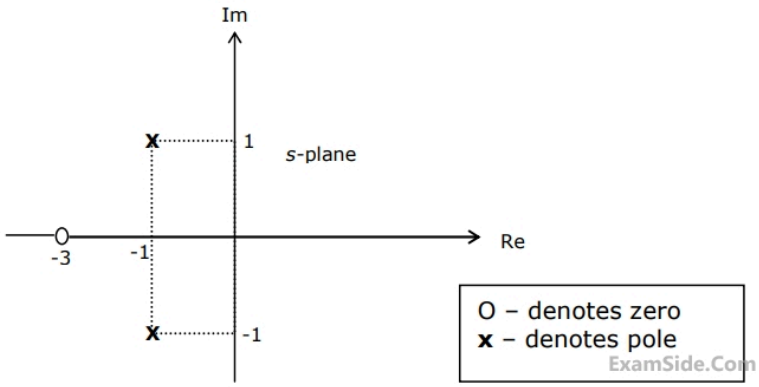
(C) Z_{11} = 3Ω and Z_{12} =0.25Ω

(D) Z_{11} = 2.25Ω and Z_{12} =0.5Ω

Q40. GATE ECE 2003

Open solution

The driving-point impedance Z(s) of a network has the pole-zero locations as shown in figure. If Z(0) = 3, then Z(s) is



(A) $\frac{3(s+3)}{s^2+2s+3}$

(B) $\frac{2(s+3)}{s^2+2s+2}$

(C) $\frac{3(s-3)}{s^2-2s-3}$

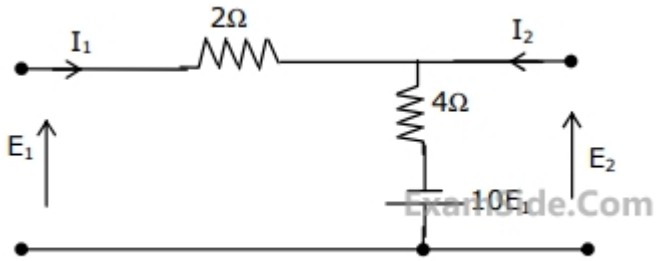
(D) $\frac{2(s-3)}{s^2-2s-3}$

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Q41. GATE ECE 2001

Open solution

The Z parameters Z_{11} and Z_{21} for the 2-port network in Fig. are



(A) $Z_{11} = -\frac{6}{11}\,\Omega;\ Z_{21} = \frac{16}{11}\,\Omega$

(B) $Z_{11} = \frac{6}{11}\,\Omega;\ Z_{21} = \frac{4}{11}\,\Omega$

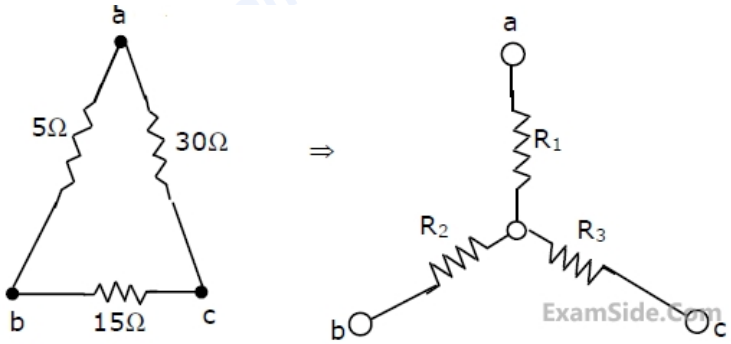
(C) $Z_{11} = \frac{6}{11}\,\Omega;\ Z_{21} = -\frac{16}{11}\,\Omega$

(D) $Z_{11} = \frac{4}{11}\,\Omega;\ Z_{21} = \frac{4}{11}\,\Omega$

Q42. GATE ECE 1999

Open solution

A Delta-connected network with its Wye-equivalent is shown in Fig.P2.4. The resistance R_1, R_2 and R_3 (in ohms) are respectively

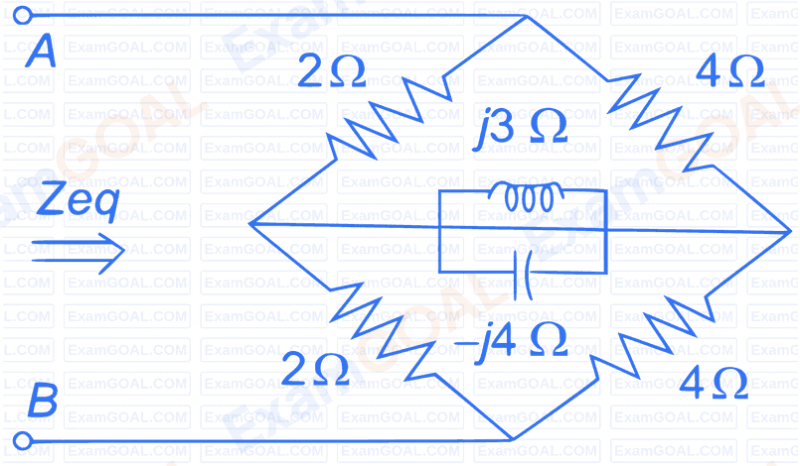


(A) 1.5, 3 and 9 (B) 3, 9 and 1.5 (C) 9, 3 and 1.5 (D) 3, 1.5 and 9

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Q43. GATE ECE 1997

In the circuit of Fig. the equivalent impedance seen across terminals A. B is



Open solution

(A) (16/3) Ω

(B) (8/3) Ω

(C) (8/3 + 12 j) Ω

(D) None of the above

Q44. GATE ECE 1995

For the 2-port network shown in Fig. determine the h-parameters. Using these parameters, calculate the output (port '2') v_2, when the output port is terminated in a 3Ω resistance and a 1 V (DC) is applied at the input port (v_1 = 1 V).

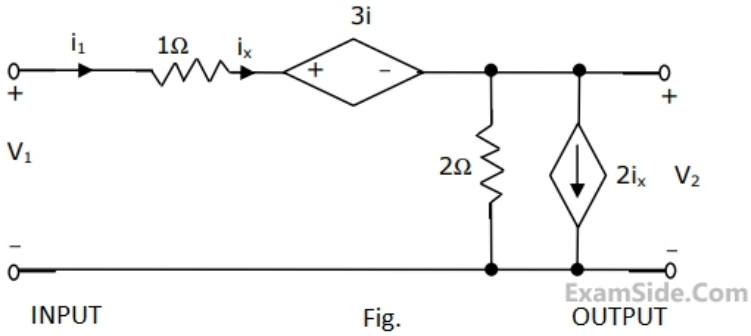


Fig.

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Open solution

Q45. GATE ECE 1992

For a 2-port network to be reciprocal

(A) $z_{11} = z_{22}$

(B) $y_{21} = y_{12}$

(C) $h_{21} = -h_{12}$

(D) AB - BC = 0

Open solution

Q46. GATE ECE 1991

Two two-port networks are connected in cascade. The combination is to be represented as a single two-port network. The parameters of the network are obtained by multiplying the individual

(A) z-parameter matrix

(B) h-parameter matrix

(C) y-parameter matrix

(D) ABCD parameter matrix

Open solution

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Q47. GATE ECE 1989

For the transfer function of a physical two-port network

(A) all the zeroes must lie only in the left half of the s-plane

(B) the poles may lie anywhere in the s-plane

(C) the poles lying on the imaginary axis must be simple

(D) a pole may lie at origin

Open solution

Q48. GATE ECE 1989

The condition AD - BC = 1 for a two-port network implies that the network is a

(A) reciprocal network (B) lumped element network

(C) lossless network (D) unilateral element network

Open solution

Q49. GATE ECE 1989

The condition AD - BC = 1 for a two-port network implies that the network is a

(A) reciprocal network (B) lumped element network

(C) lossless network (D) unilateral element network

Open solution

Q50. GATE ECE 1988

Two two-port network A and B are connected in parallel. The combination is to be represented as a single two-port network. The parameters of this network are obtained by addition of the individual

(A) z parameters (B) h parameters (C) y parameters (D) ABCD parameters

Open solution

Marks 5 8 questions

Q51. GATE ECE 2003

Find the Y-parameters (short ciruit admittance parameters) for the network shown in figure.

Open solution

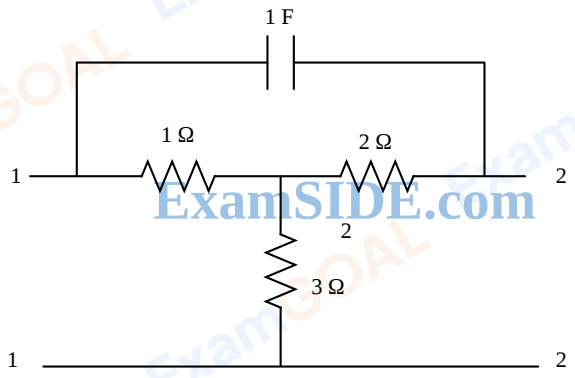


Fig.

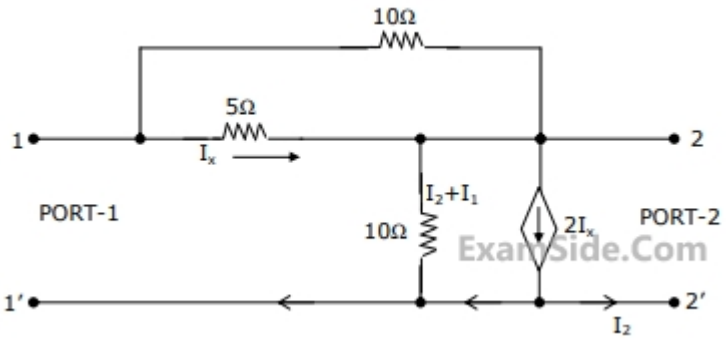
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Q52. GATE ECE 2002

Open solution

Consider the network in Fig.
(a) Find its short-circuit admittance parameters.
(b) Find the open-ciruit impedance Z_{22}.



Q53. GATE ECE 2001

Open solution

The admittance parameters of a 2-port network shown in Fig. are given by $Y_{11} = 2\ mho, Y_{12}$ = - 0.5 mho, Y_{21} = 4.8 mho, Y_{22} = 1 mho. The output port is terminated with a load admittance Y_L = 0.2 mho. Find E_2 for each of the following conditions:
(a) $E_1 = 10\angle 0^\circ\ V$
(b) $I_1 = 10\angle 0^\circ\ A$
(c) A source $10\angle 0^\circ\ V$ in series with a 0.25 Ω resistor is connected to the input port.

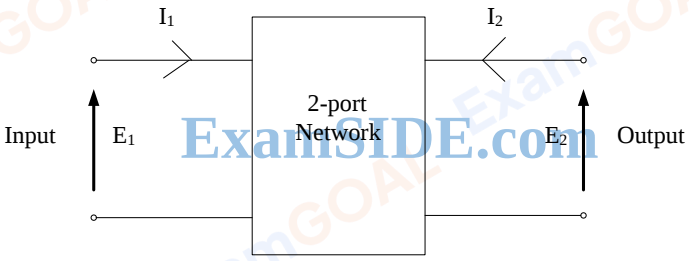
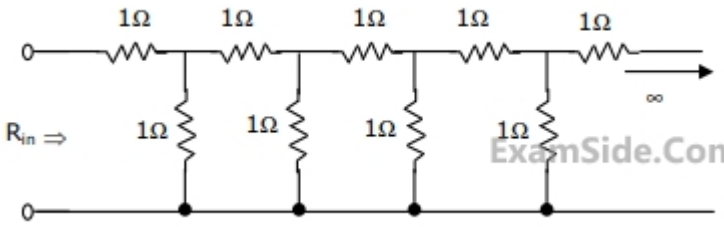


Fig.

Q54. GATE ECE 1996

Open solution

Find the input resistance R_{in} of the infinite section resistive network shown in Fig.



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Q55. GATE ECE 1996

Open solution

The open circuit impedance matrix Z_{OC} of a three-terminal two-port network with A as the input terminal, B as the output terminal, and C as the common terminal, is given as

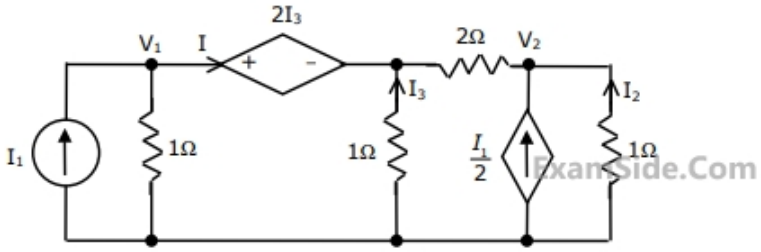
$$[Z_{OC}] = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix}$$

Write down the short circuit admittance matrix Y_{SC} of the network viewed as a two-port network, but now taking B as the input terminal, C as the output terminal and A as the common terminal.

Q56. GATE ECE 1995

Open solution

Find the current-transfer-ratio, I_2/I_1, for the network shown below (Fig). Also, mark all branch currents.

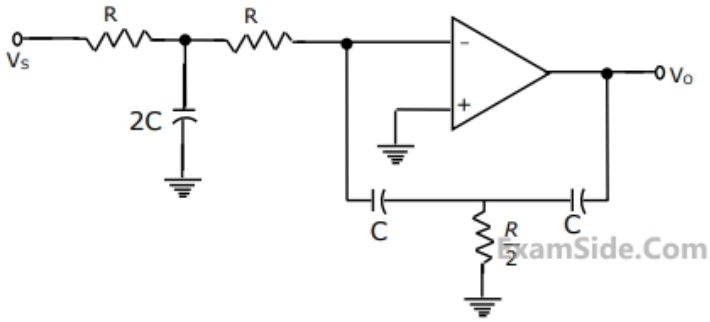


Q57. GATE ECE 1995

Open solution

Show that the system shown in Fig. is a double integator. In other words, prove that the transfer gain is given by

$\frac{V_0(s)}{V_s(s)} = -\frac{1}{(CR\,s)^2}$, assume ideal OP-Amp



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Q58. GATE ECE 1994

Assuming that the amplifier shown in the Fig., below, is a voltage-controlled voltage source, show that the voltage transfer function of the network is given by
$T(s) = \frac{V_2(s)}{V_1(s)} = \frac{K}{s^2 + (3-K)s + 1}$

Open solution

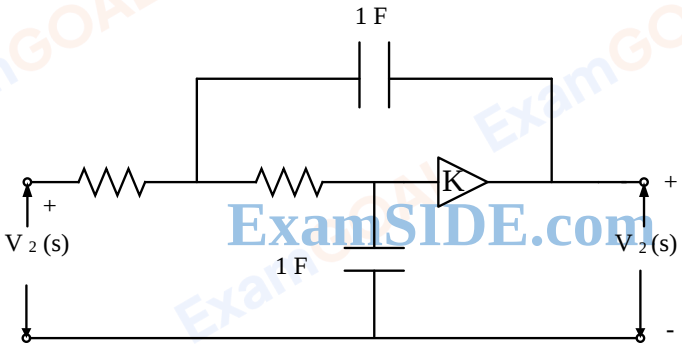


Fig.

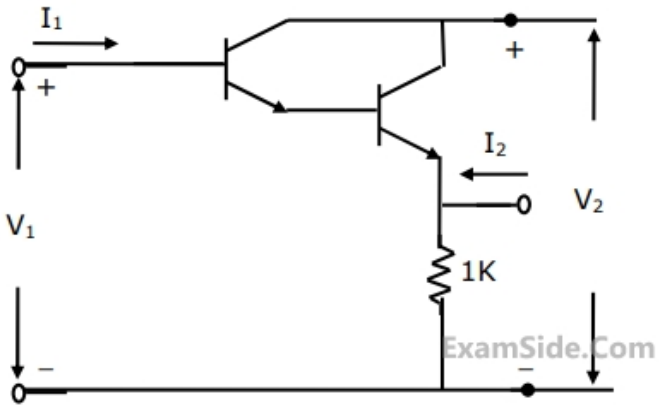
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Marks 8 1 questions

Q59. GATE ECE 1992

The two-port Darlington impedance booster of figure uses identical transistors $(h_{ie} = 1\,k,\ h_{fe} = 100,\ h_{re} = h_{oe} = 0$. Calculate the z-parameters of the network. (Use relavant approximations)

Open solution



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Answer Key

Marks 1

1. **80**	2. **A**	3. **48**	4. **4.3 to 5.3**	5. **3**
6. **D**	7. **A**	8. **A**	9. **C**	10. **B**
11. **C**	12. **A**	13. **A**	14. **B**	15. **AD-BC = 1**
16. **A**				

Marks 2

17. **D**	18. **A**	19. **1.5**	20. *Scan QR*	21. **A**
22. **4**	23. **0.8**	24. **A**	25. **B**	26. **C**
27. **2.6**	28. **1.25**	29. **D**	30. **D**	31. **C**
32. **A**	33. **B**	34. **D**	35. **C**	36. **D**
37. **D**	38. **D**	39. **A**	40. **B**	41. **C**
42. **D**	43. **D, B**	44. V_2 **= 6 V**	45. **B, C**	46. **D**
47. **D, C**	48. **A**	49. **A**	50. **C**	

Marks 5

51. **solve it**	52. **(a)** $\begin{bmatrix} 0.3 & -0.3 \\ 0.1 & 0 \end{bmatrix}$ **(b)** $Z_{22} = 10$	53. **(a)** E_2 **= -40 V** **(b)** E_2 **= - 10 V** **(c)** E_2 **= - 20 V**	54. $R_{in} = \frac{V5+1}{2}\,\Omega$	55. $Y_{SC} = \begin{bmatrix} -2 & -1 \\ -3 & -1 \end{bmatrix}$
56. $\frac{I_2}{I_1} = -\frac{11}{26}$	57. **solve it**	58. **solve it**		

Marks 8

59. $Z_{11} = 10.1\ M\,\Omega$ $Z_{12} = 1\ K\,\Omega$ $Z_{21} = 10\ M\,\Omega$ $Z_{22} = 1\ K\,\Omega$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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